

Week 1 Homework Review

Scholars High School Physics 1: Mechanics

Writing problem: Technical 5.5 / 7, Style 1 / 1 · Challenge problems: 10 / 10, one retry

The headline

Your physics was right. Every conceptual step in the writing problem was correct, and the grader said so. The marks came off for *bookkeeping*: an assumed scale, an unchecked formula, and a missing sanity check. That is good news — these are habits, and habits are far quicker to fix than understanding.

1 What you got right

What you got right

- **(a) Direction is unknowable.** A dot diagram records positions, not the order they were visited. You said so and explained why. Full credit.
- **(b) The speed increases.** You compared distances covered in equal time intervals — exactly the right reading of a dot diagram — and drew the right conclusion.
- **(d) The centred-interval technique.** To estimate the speed *at* the second-to-last dot you used the dots on *either side* of it rather than one adjacent gap. That is the correct method, and a lot of students get it wrong. The grader specifically approved the choice of dots and the elapsed time.

2 What went wrong

1. You invented a scale instead of deriving one

This is the big one, and it caused both numerical errors.

You read the dot positions off the drawing as

0, 1, 4, 9, 16, 25, 36

and then wrote “*given that position n and position $n + 1$ are one meter apart.*” Those numbers are real, but they are **coordinates of the drawing**, not metres. Nothing in the problem said one coordinate unit was a metre — that was an assumption, and it was wrong by a factor of nearly three hundred.

Where the metres had to come from

The diagram was printed at a stated scale of 1 : 1, so a ruler laid on it measures real distances. The last gap spans $36 - 25 = 11$ coordinate units and measures 3.9 cm. That one measurement fixes everything:

$$1 \text{ coordinate unit} = \frac{3.9 \text{ cm}}{11} = 0.35 \text{ cm} = 3.5 \text{ mm}.$$

With the scale in hand, the two answers come out as follows.

	your answer	correct answer
(c) average speed	$\frac{36}{6} = 6 \text{ m/s}$	$\frac{36 \times 0.35 \text{ cm}}{6 \text{ s}} = 2.1 \text{ cm/s}$
(d) instantaneous speed	$\frac{36 - 16}{7 - 5} = 10 \text{ m/s}$	$\frac{20 \times 0.35 \text{ cm}}{2 \text{ s}} = 3.5 \text{ cm/s}$

Notice that both answers are too large by the *same* factor of 282. That is the signature of one bad conversion, not two separate mistakes — and it is why fixing the single habit fixes both parts.

2. You wrote the formula upside down

In part (b) you wrote:

“distance = speed / time” instead of distance = average speed $\times$ time

The calculations that followed used d/t correctly, so the wrong formula never actually got used — which is precisely why it survived to the page. A reader can only judge what you wrote, and the next time you reach for that formula on a problem where the right answer is not obvious, you will follow the words.

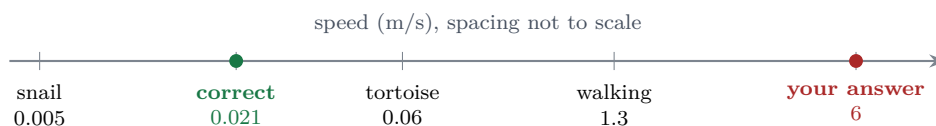
Check a rearrangement with units, every time

Substitute units for quantities and see whether both sides match. It takes five seconds and needs no physics:

$$\frac{\text{m/s}}{\text{s}} = \text{m/s}^2 \quad (\text{an acceleration — not a distance}), \quad \text{m/s} \times \text{s} = \text{m} \quad \checkmark$$

3. No sanity check

6 m/s is a fast run — roughly 22 km/h. The diagram is a few centimetres across and covers six seconds. Those two facts cannot both be true, and noticing the clash costs nothing.



4. A phrasing slip

You wrote “the *instantaneous speed* occurs between the 5th dot and the 7th dot.” It does not occur between anything. Instantaneous speed is the speed **at one instant**; what you computed was an average over the surrounding interval, which **estimates** the instantaneous speed **at the middle dot**. The method was right — only the sentence was loose. Say it as:

“The average speed from dot 5 to dot 7 estimates the instantaneous speed *at* dot 6, because the interval is centred on that dot.”

3 The same pattern showed up again

Week 2, Problem 2 (notation matching, 5/7 after a retry): your first attempt swapped v and v_x . Not a concept error — a subscript error. Same family as everything above.

symbol	name	sign	read it as
x	coordinate along x	either	“where it is, relative to the origin”
Δx	displacement along x	either	“how far and which way it moved”
v_x	velocity along x	either	“how fast and which way it is going”
v	speed	never negative	“how fast it is going”

The rule in one line: a subscript names an axis and keeps the sign; a bare v means speed and never carries one.

4 The knowledge point

Numbers read off a diagram are in the diagram's units

Read → Convert → Divide → Check.

1. **Read** the positions in the diagram's own units, and *name* them: write “36 squares,” never a bare “36.” A number with its unit attached makes the missing conversion visible, because squares/s is not a speed.
2. **Convert** using a scale you *measured or were given*: one length in the diagram whose true size you know. If you cannot point at that measurement, you assumed a scale rather than found one.
3. **Divide** by the elapsed time — and count intervals, not dots.
4. **Check** by converting into units you have a feel for before writing the answer down.

Three questions, thirty seconds, before you submit

1. *Have I named the units at every step?* A bare number is an invitation to divide it by seconds and call the result m/s.
2. *Where did my metres come from?* Trace them back to a real ruler measurement or a given scale. No trace, no metres.
3. *Does the size make sense?* Convert into something you can picture and compare it with a thing you know.

5 Where this now lives in the book

Chapter 1 of *Physics 1: Mechanics* was rewritten around these four points (46 → 53 pages). **§1.3**, “**Grid units are not metres**” is new: the three-step rule, a dot diagram on a grid with the ruler measurement marked, a worked example on fresh numbers, and a modelling note, “Three habits that catch this error.” **§1.4** adds “Check a rearrangement with its units”; **§1.5** the centred

estimate $v_n \approx (x_{n+1} - x_{n-1})/2\tau$ and the phrasing point; **§1.9** the symbols table above. Plus 8 new exercises and 2 new check-your-understanding boxes, all answered.

Sources: your graded Week 1 homework and Week 2 Problem 2, AoPS course 4926.